Vietnamese German University

Department of Computer Science and Engineering

COURSE PROJECT REPORT

Course Title: [Real-Time Systems / RTOS]

Smart Controller System using RTOS

Group Members

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Project Requirements (Reference)

Objective: Develop an RTOS-based Smart Climate Control System that monitors temperature and humidity, and controls a heater, cooler, and humidifier (represented by LEDs) to maintain target conditions. The project is a course demo and need not reflect real-world physical operation.

Hardware: YoluUNO (ESP32-S3) microcontroller; DHT20 temperature/humidity sensor on I2C; three two-color LED actuators representing Heater (D3/D4), Cooler (D5/D6), and Humidifier (D7/D8); onboard status LED on GPIO48.

RTOS: FreeRTOS (or any RTOS suitable for the microcontroller), using the task scheduler.

Required behaviors: Serial print of temperature/humidity every 5 seconds; onboard LED blinks every 1 second; Heater LED shows GREEN/ORANGE/RED according to 3 defined temperature thresholds; Cooler activates for a fixed duration (e.g., 5 s) when temperature exceeds a threshold, then re-checks; Humidifier runs a GREEN (5 s) → YELLOW (3 s) → RED (2 s) sequence when humidity falls below a threshold, then re-checks humidity to decide whether to repeat.

Required tasks: Blinky Task, Read Temperature Task, Cooler Task, Heater Task, Humidifier Task.

Deliverable: GitHub repository link and result snapshots.

1. Introduction

2. System Design using State Machine

3. System Implementation

4. Validation Results in Simulation

4.1 Monitoring Output

Figure 1 shows the LCD displaying the latest temperature and humidity values obtained from the DHT20 sensor. The monitoring task updates the measurements every 5 seconds and forwards the data to the Decision Maker Task for evaluation.

`images/monitoring_lcd.png`

Figure 1: LCD displaying the latest temperature and humidity measurements.

4.2 Heater Behavior

Figure 2 demonstrates the Heater indicator state transitions. The LED displays GREEN when the temperature is within the normal range (20–25°C), YELLOW when the temperature enters the warning range (15–20°C or 25–30°C), and RED when the temperature reaches a critical condition ($<$15°C or $>$30°C).

4.3 Cooler Behavior

Figure 3 shows the Cooler Task activation. When the temperature exceeds 28°C, the cooler LED turns GREEN and remains active for 5 seconds. After the cooling interval expires, the LED turns OFF and the temperature is evaluated again.

4.4 Humidifier Behavior

The Humidifier control sequence is described below. When humidity falls below 40%, the humidifier executes a timed sequence consisting of GREEN (5 s), YELLOW (3 s), and RED (2 s). After completion, images/heater_critical_red.png

(a) Critical RED state.

images/heater_safe_green.png

(b) Safe GREEN state.

images/heater_warning_yellow.png

(c) Warning YELLOW state.

Figure 2: Heater indicator behavior for critical, safe, and warning temperature ranges.

images/cooler_active_green.png

(a) Cooler active in the GREEN state.

images/cooler_idle_off.png

(b) Cooler inactive in the OFF state.

Figure 3: Cooler indicator during active cooling and after returning to the idle state.

the system returns to the idle state and rechecks the humidity condition.

4.5 Source Code Repository

GitHub link: https://github.com/Khainh2005/RTOSproject

5. Conclusion and Perspective